

Assessment Demographic Data • Chief complaint History of Present illness including functional limitation Past Medical Surgical history Family history Occupational and Personal history Socio Economic Environmental and Psychological History Drug and Treatment History etc --- **Musculoskeletal Disorders** • Vital parameters Temperature Respiratory Rate Peripheral Pulses Blood Pressure • Body Built of Patient Artitude of Limb Bony contours Deformities Trophic changes • Examination of Posture • Examination of Spasm Tenderness Edema /swelling/wasting including Limb girth measurement • Scar examination • **Motor Examination** Range of Motion Joint Play and End Feel Muscle Strength Power Endurance Flexibility Isometric Resisted Testing Limb length discrepancy Pelvic inclination Examination of Balance Gait and External aids/assistive Device used etc • **Sensory Examination** Superficial and Deep sensation Reflexes Dermatomes and Myotomes etc

- **Dermatomes** • **Definition** Areas of skin supplied by sensory fibers from a single spinal nerve root • **Purpose** Helps identify sensory loss due to nerve root lesions • **Examples** • C6 Thumb • C7 Middle finger • C8 Little finger • L4 Medial side of leg • L5 Dorsum of foot and great toe • S1 Lateral side of foot • **Normal** Sensation is intact in the tested dermatome • **5 Myotomes** • **Definition** Groups of muscles predominantly supplied by a single spinal nerve root • **Purpose** Assesses motor function of spinal nerve roots • **Examples** • C5 Shoulder abduction • C6 Wrist extension • C7 Elbow extension • C8 Finger flexion • L2 Hip flexion • L3 Knee extension • L4 Ankle dorsiflexion • L5 Great toe extension • S1 Ankle plantarflexion
- **Normal** Full strength usually 5/5 on MMT --- **Assessment of Hand Function** • **1 Hand Pinches** • **Definition** Holding an object between the thumb and fingers • **Types** Tip to tip lateral pinch three jaw chuck pinch • **Assessment** Pinch gauge
- **Example** Holding a key or picking up a coin • **2 Hand Grip** • **Definition** Ability to grasp and hold objects using the hand
- **Types** Power grip and precision grip • **Assessment** Hand dynamometer • **Normal** Adequate grip strength for daily activities • **Example** Holding a bottle or squeezing a ball • **3 Routine Sensory Motor Evaluation** • **Definition** Assessment of sensory and motor functions of the hand • **Motor assessment** Muscle strength grip pinch thumb opposition and finger movements • **Sensory assessment** Light touch pain temperature vibration and joint position sense • **Normal** Intact sensation normal strength and coordinated movements • **Example** Weak thumb opposition may indicate median nerve involvement • **4 Stereognosis** • **Definition** Ability to identify an object by touch without seeing it • **Assessment** Patient identifies an object with eyes closed • **Normal** Correct identification of objects • **Example** Identifying a coin or key by touch
- **Abnormal** Inability to recognize an object by touch is called astereognosis --- // // // // **Cardiovascular and Pulmonary dysfunction** • Symmetry/Deformities of Head Neck Extremities Trophic changes • Examination of cough sputum tactile and vocal fremitus mobility of thoracic spine and rib cage percussion etc • **1 Auscultation** • **Definition** Listening to internal body sounds using a stethoscope • **Normal heart sounds** • S1 Closure of mitral and tricuspid valves heard as lub • S2 Closure of aortic and pulmonary valves heard as dub • **Normal heart rate** 60 to 100 beats/minute in adults • **Abnormal heart sounds** • Murmur Abnormal sound due to turbulent blood flow • S3 May be normal in young adults but can indicate heart failure • S4 May occur with reduced ventricular compliance • **Normal breath sounds** • Vesicular Soft low pitched sounds over most lung fields • Bronchial Heard normally over the trachea • Bronchovesicular Heard near the major bronchi
- **Abnormal breath sounds** • Crackles Popping sounds eg pulmonary edema or pneumonia • Wheeze Musical sound eg asthma • Rhonchi Low pitched sounds due to airway secretions • Stridor Harsh sound due to upper airway obstruction
- **Reduced or absent breath sounds** May occur with pneumothorax or pleural effusion • **2 Chest Expansion** • **Definition** Measurement of the increase in chest circumference during inspiration • **Method** Measure chest circumference during maximal expiration and inspiration using a measuring tape • **Common sites** Axillary nipple line and xiphisternal level • **Normal** Approximately 3 to 5 cm in healthy adults depending on age and measurement site
- **Example** Reduced chest expansion in COPD • **3 Symmetry of Chest Movement** • **Definition** Assessment of equal movement of both sides of the chest during breathing • **Method** Observe chest movement or place hands symmetrically over the chest • **Normal** Both sides expand equally during inspiration • **Abnormal** Unequal chest expansion
- **Example** Reduced movement on one side in pleural effusion or pneumothorax • **4 Body Mass Index BMI** • **Definition** Measurement of body weight in relation to height • **Formula** BMI equals Weight kg divided by Height m²
- **Normal adult BMI WHO** 18 point 5 to 24 point 9 kg/m² • **Example** Weight 60 kg height 1 point 65 m BMI equals 22 point 0 kg/m² • **5 Waist Hip Ratio WHR** • **Definition** Ratio of waist circumference to hip circumference • **Formula** WHR equals Waist circumference divided by Hip circumference • **Normal reference values WHO** • Men less than or equal to 0 point 90
- Women less than or equal to 0 point 85 • **Example** Waist 80 cm hip 100 cm WHR equals 0 point 80 • **Purpose** Assesses abdominal fat distribution and associated health risk

3 Special Test: Cervical Spine: Test for Vertigo & Dizziness tests: Vertebral Artery Test Dizziness Test etc. Test for Thoracic Outlet Syndrome: Adson's test, Roos stress Test etc. Test for Neurological symptoms: Foraminal Compression Test, Distraction Test, Shoulder Depression Test etc. PFDP 3 Cervical Special Tests .pdf **CERVICAL SPECIAL TESTS** --- **1 Spurlings Test /Foraminal Compression Test/ PURPOSE** To assess cervical nerve root compression /cervical radiculopathy/ **PROCEDURE** • Patient sits comfortably • Extend laterally flex and rotate the head towards the affected side

- Apply gentle downward compression on the head **POSITIVE SIGN** Radiating pain numbness or tingling into the upper

limb **INDICATES** • Cervical nerve root compression • Cervical radiculopathy --- **2 Cervical Distraction Test PURPOSE** To assess cervical nerve root compression **PROCEDURE** • Patient lies supine • Therapist gently grasps the head and applies an upward traction force **POSITIVE SIGN** Relief of neck or arm pain **INDICATES** • Cervical radiculopathy • Foraminal compression --- **3 Upper Limb Tension Test /ULTT/ PURPOSE** To assess neural tissue mobility and detect peripheral nerve or cervical nerve root involvement --- **ULTT 1 /MEDIAN NERVE BIAS/ Procedure** • Depress the shoulder • Abduct the shoulder to 110 • Supinate the forearm • Extend the wrist and fingers • Externally rotate the shoulder • Slowly extend the elbow • Finally side flex the neck away if required **Positive Sign** • Pain tingling or numbness along the median nerve distribution • Difference in symptoms compared to the opposite side **Indicates** • Median nerve tension • Cervical radiculopathy --- **ULTT 2A /MEDIAN NERVE BIAS/ Procedure** • Shoulder depression • Elbow extension • External rotation of shoulder • Forearm supination • Wrist and finger extension **Positive Sign** Reproduction of median nerve symptoms **Indicates** Median nerve irritation --- **ULTT 2B /RADIAL NERVE BIAS/ Procedure** • Shoulder depression • Internal rotation of shoulder • Elbow extension • Forearm pronation • Wrist finger and thumb flexion with ulnar deviation **Positive Sign** Pain or tingling along the dorsolateral forearm and hand **Indicates** Radial nerve tension --- **ULTT 3 /ULNAR NERVE BIAS/ Procedure** • Shoulder depression • Shoulder abduction and external rotation • Elbow flexion • Forearm pronation • Wrist extension with radial deviation • Finger extension **Positive Sign** Pain or tingling along the medial forearm and little finger **Indicates** Ulnar nerve tension --- **4 Vertebral Artery Test PURPOSE** To assess vertebrobasilar insufficiency before cervical mobilization or manipulation **PROCEDURE** • Patient lies supine • Therapist extends rotates and laterally flexes the neck • Hold for about 30 seconds while observing symptoms **POSITIVE SIGN** • Dizziness • Vertigo • Nystagmus • Blurred vision • Nausea **INDICATES** Vertebrobasilar insufficiency **Precaution** Stop the test immediately if symptoms appear --- **SPECIAL EXAMINATION TESTS** --- **1 DIZZINESS TEST PURPOSE** To assess for cervical vascular compromise/vertebrobasilar insufficiency that may produce dizziness during cervical movement **PROCEDURE** • Patient is positioned sitting • Examiner stabilizes the patients head • Slowly place the cervical spine into extension and rotation • Observe the patient for symptoms **POSITIVE SIGN** • Dizziness • Vertigo • Nausea • Visual disturbance • Nystagmus **INDICATES** Possible vertebrobasilar insufficiency and requires further medical evaluation --- **2 SHOULDER DEPRESSION TEST PURPOSE** To assess nerve root/nerve tissue irritation or shortening of neural structures in the cervical and upper limb region **PROCEDURE** • Patient is sitting • Examiner stabilizes/depresses the shoulder on the tested side • Passively side flex the patients neck away from the tested side • The position places tension on the cervical nerve structures **POSITIVE SIGN** • Reproduction of the patients radiating pain/paresthesia into the upper limb **INDICATES** Possible cervical nerve root or brachial plexus involvement --- **3 PINCH GRIP TEST PURPOSE** To assess the function of the anterior interosseous nerve /AIN/ and flexor pollicis longus/index FDP function **PROCEDURE** • Ask the patient to make a pinch/OK sign by touching the tip of the thumb to the tip of the index finger • Observe the shape of the circle formed **POSITIVE SIGN** • Patient cannot make a proper round OK circle • Instead the pinch becomes flattened/pulp to pulp **INDICATES** Anterior interosseous nerve palsy/dysfunction --- **4 CARPAL COMPRESSION TEST PURPOSE** To assess for carpal tunnel syndrome due to median nerve compression **PROCEDURE** • Patients wrist is in a neutral position • Examiner applies direct pressure over the carpal tunnel with the thumbs/fingers • Maintain pressure for approximately 30 seconds **POSITIVE SIGN** • Reproduction of • Pain • Tingling • Numbness • Paresthesia in the median nerve distribution **INDICATES** Carpal Tunnel Syndrome / median nerve compression --- **5 HAND ELEVATION TEST PURPOSE** To assess for carpal tunnel syndrome/median nerve compression **PROCEDURE** • Patient raises both hands above the head • Maintain the elevated position for approximately 1 2 minutes • Ask the patient about symptoms **POSITIVE SIGN** • Reproduction of pain numbness or tingling in the median nerve distribution **INDICATES** Possible Carpal Tunnel Syndrome --- **6 FROMENTS SIGN PURPOSE** To assess the ulnar nerve particularly the function of adductor pollicis **PROCEDURE** • Ask the patient to hold a piece of paper between the thumb and index finger • Examiner attempts to pull the paper away **POSITIVE SIGN** Patient compensates by • Flexing the IP joint of the thumb using flexor pollicis longus **INDICATES** Ulnar nerve palsy/weakness of adductor pollicis4 Special Test: Shoulder Joint: Test for Shoulder Instability: Apprehension, Jerk, Drawer, Sulcus sign etc. Test for Impingement: Hawkins-Kennedy Test Neer's Impingement Test, Painful Arc Test, Empty Can Test, Drop Arm Test etc. Test for Labral Lesion: Active Compression test of O'Brien, KIM test etc. Test for Scapular Dyskinesia: Scapular Dyskinesia test,

Scapular Load Test etc. Test for Muscle Pathology: Yergason's, Speed's, Drop-Arm, Empty can, Lift-Off Sign, Test for Supraspinatus, Infraspinatus, Horn blower Sign etc. PFDP 4 Shoulder Special Tests.pdf --- **I INSTABILITY TESTS** --- **1 APPREHENSION TEST Purpose** To assess anterior shoulder instability **Procedure** Abduct the shoulder to 90 flex the elbow to 90 and slowly externally rotate the shoulder **Positive Sign** Patient becomes apprehensive or fears shoulder dislocation **Indicates** Anterior glenohumeral instability--- **2 JERK TEST Purpose** To assess posterior shoulder instability and posterior labral lesions **Procedure** Shoulder abducted to 90 and internally rotated **Apply** axial compression through the elbow while horizontally adducting the arm **Positive Sign** Painful clunk or jerk **Indicates** Posterior instability or posterior labral tear--- **3 DRAWER TEST Purpose** To assess anterior and posterior glenohumeral instability **Procedure** Stabilize the scapula and

translate the humeral head anteriorly and posteriorly Positive Sign Excessive humeral head movement compared to the opposite side Indicates Shoulder instability--- **4 SULCUS SIGN** Purpose To assess inferior glenohumeral instability Procedure Apply downward traction to the relaxed arm Positive Sign Visible sulcus below the acromion Indicates Inferior or multidirectional shoulder instability--- **II IMPINGEMENT TESTS** --- **1 HAWKINS KENNEDY TEST** Purpose Assess subacromial impingement Procedure Flex shoulder and elbow to 90 and internally rotate the shoulder Positive Sign Pain Indicates Subacromial impingement--- **2 NEERS IMPINGEMENT TEST** Purpose Assess subacromial impingement Procedure Stabilize the scapula and passively flex the internally rotated arm Positive Sign Pain Indicates Subacromial impingement--- **3 PAINFUL ARC TEST** Purpose Assess shoulder impingement Procedure Ask the patient to actively abduct the shoulder Positive Sign Pain between 60 120 of abduction Indicates Subacromial impingement--- **4 EMPTY CAN JOBES TEST** Purpose Assess supraspinatus tendon Procedure Shoulder abducted to 90 in the scapular plane internally rotated thumb down resist downward pressure Positive Sign Pain or weakness Indicates Supraspinatus tendinopathy or tear--- **5 DROP ARM TEST** Purpose Assess rotator cuff tear Procedure Passively abduct the arm and ask the patient to slowly lower it Positive Sign Arm drops suddenly or cannot be lowered smoothly Indicates Supraspinatus tear/Rotator cuff tear--- **III LABRAL LESION TESTS** --- **1 ACTIVE COMPRESSION TEST OBRIEN TEST** Purpose Assess SLAP lesion and AC joint pathology Procedure Shoulder flexed to 90 adducted 10 15 Thumb pointing downward Apply downward resistance Repeat with palm facing upward Positive Sign Pain or clicking that decreases with the palm up position Indicates SLAP lesion or AC joint pathology- **2 KIM TEST** Purpose Assess posteroinferior labral lesion Procedure Shoulder abducted to 90 Apply axial compression with upward and posterior force while elevating the arm diagonally Positive Sign Posterior shoulder pain Indicates Posteroinferior labral tear--- **IV SCAPULAR DYSKINESIA TESTS** --- **1 SCAPULAR DYSKINESIS TEST** Purpose Assess abnormal scapular movement Procedure Patient performs repeated shoulder flexion and abduction with light weights while the examiner observes scapular motion Positive Sign Scapular winging or asymmetrical movement Indicates Scapular dyskinesia--- **2 SCAPULAR LOAD TEST SCAPULAR ASSISTANCE TEST** Purpose Assess the effect of scapular assistance on shoulder movement Procedure Examiner manually assists upward rotation and posterior tilt of the scapula during arm elevation Positive Sign Pain decreases or shoulder movement improves Indicates Scapular dyskinesia--- **V MUSCLE PATHOLOGY TESTS** --- **1 YERGASONS TEST** Purpose Assess biceps tendon pathology Procedure Patient supinates the forearm against resistance with the elbow flexed to 90 Positive Sign Pain or snapping over the bicipital groove Indicates Biceps tendinitis or transverse humeral ligament injury--- **2 SPEEDS TEST** Purpose Assess long head of biceps tendon Procedure Shoulder flexed to 90 elbow extended forearm supinated resist downward pressure Positive Sign Pain in the bicipital groove Indicates Biceps tendinitis or SLAP lesion--- **3 LIFT OFF TEST** Purpose Assess Subscapularis muscle Procedure Place the dorsum of the hand on the lower back and ask the patient to lift the hand away Positive Sign Unable to lift the hand or marked weakness Indicates Subscapularis tear or weakness--- **4 TEST FOR SUPRASPINATUS** Purpose Assess supraspinatus muscle Procedure Perform the Empty Can Test against resistance Positive Sign Pain or weakness Indicates Supraspinatus pathology--- **5 TEST FOR INFRASPINATUS** Purpose Assess infraspinatus muscle Procedure Patient externally rotates the shoulder against resistance with the elbow flexed to 90 Positive Sign Pain or weakness Indicates Infraspinatus tear or weakness--- **6 HORNBLLOWERS SIGN** Purpose Assess Teres Minor function Procedure Shoulder elevated to 90 in the scapular plane with elbow flexed to 90 Ask the patient to maintain external rotation Positive Sign Unable to maintain the position or external rotation Indicates Teres minor tear or severe rotator cuff pathology // **5 Special Test:** Elbow Joint: Test for ligament instability: Valgus- Varus ligament Test etc. Test for Tennis Elbow: Cozens, Mill's, Maudsley's etc. Test for Golfer's Elbow. Test for Neurological symptoms: Pinch Grip Test, Tinel Sign etc. PFDP 5 Elbow Special Tests.pdf --- **I TESTS FOR LATERAL EPICONDYLITIS TENNIS ELBOW** --- **1 COZENS TEST** Purpose To assess lateral epicondylitis Tennis Elbow Procedure • Patient makes a fist • Elbow flexed to 90 • Forearm pronated • Wrist extended and radially deviated against resistance Positive Sign Pain over the lateral epicondyle Indicates Lateral epicondylitis --- **2 MILLS TEST** Purpose To assess lateral epicondylitis Procedure • Patient sits with the elbow extended • Forearm pronated • Therapist passively flexes the wrist while maintaining elbow extension Positive Sign Pain over the lateral epicondyle indicates Lateral epicondylitis --- **3 MAUDSLEYS TEST MIDDLE FINGER TEST** Purpose To assess lateral epicondylitis especially involvement of the Extensor Digitorum Communis EDC Procedure • Elbow extended • Forearm pronated • Patient resists extension of the middle finger Positive Sign Pain over the lateral epicondyle Indicates Lateral epicondylitis --- **II TEST FOR MEDIAL EPICONDYLITIS GOLFERS ELBOW** --- **4 GOLFERS ELBOW TEST MEDIAL EPICONDYLITIS TEST** Purpose To assess medial epicondylitis Golfers Elbow Procedure • Patient sits with the elbow extended • Forearm supinated • Therapist passively extends the wrist and fingers while maintaining elbow extension Positive Sign Pain over the medial epicondyle Indicates Medial Epicondylitis Golfers Elbow --- **III NERVE COMPRESSION TEST** --- **5 TINEL SIGN CUBITAL TUNNEL** Purpose To assess ulnar nerve irritation or compression Procedure • Slightly flex the elbow • Tap gently over the ulnar nerve behind the medial epicondyle Positive Sign Tingling or electric shock sensation along the ring and little fingers Indicates Cubital Tunnel Syndrome Ulnar nerve entrapment --- **IV LIGAMENT STABILITY TESTS** --- **6 VALGUS STRESS TEST** Purpose To assess the Medial Ulnar Collateral

Ligament MCL/UCL Procedure • Elbow slightly flexed 20 30 • Stabilize the humerus • Apply a valgus force to the forearm Positive Sign Pain or excessive medial joint opening Indicates Medial collateral ligament injury -- - **7 VARUS STRESS TEST** Purpose To assess the Lateral Collateral Ligament LCL Procedure • Elbow slightly flexed 20 30 • Stabilize the humerus • Apply a varus force to the forearm Positive Sign Pain or excessive lateral joint opening Indicates Lateral collateral ligament injury // // // // **6 Special Test: Wrist & Hand Complex:** Test for Carpal Tunnel Syndrome: Carpal Compression test, Hand Elevation Test, Phalen's, Froment's sign etc. Test for Bone, Ligament, Capsule & joint Instability: Ulnar fovea Test, Ulnar Menisco Triquetral dorsal Glide Test, Watson Test, Bunnel-Littler Test etc. Test for circulation & Swelling: Allen Test, Digital blood flow, Figure of 8 measurement of swelling. Hand volume Test etc. PFDP 6 Wrist Special Tests .pdf --- **WRIST SPECIAL TESTS** --- **I CARPAL TUNNEL SYNDROME TESTS** --- **1 PHALENS TEST** Purpose To assess Carpal Tunnel Syndrome Median nerve compression Procedure • Ask the patient to flex both wrists maximally • Dorsum of both hands are pressed together • Hold for 60 seconds Positive Sign Pain tingling or numbness in the thumb index middle and lateral half of ring finger Indicates Carpal Tunnel Syndrome --- **2 REVERSE PHALENS TEST PRAYER TEST** Purpose To assess Carpal Tunnel Syndrome Procedure • Ask the patient to press both palms together • Maintain maximum wrist extension for 60 seconds Positive Sign Pain or paresthesia in the median nerve distribution Indicates Carpal Tunnel Syndrome --- **3 TINEL'S SIGN WRIST** Purpose To assess Median nerve irritation Procedure • Tap gently over the median nerve at the carpal tunnel Positive Sign Tingling or electric shock sensation in the median nerve distribution Indicates Carpal Tunnel Syndrome --- **II TENDON PATHOLOGY TEST** --- **4 FINKELSTEIN'S TEST** Purpose To assess De Quervain's Tenosynovitis Procedure • Patient places the thumb inside the fist • Fingers wrap around the thumb • Examiner passively deviates the wrist towards the ulnar side Positive Sign Pain over the radial styloid Indicates De Quervain's Tenosynovitis --- **III LIGAMENT STABILITY TEST** --- **5 WATSON SCAPHOID SHIFT TEST** Purpose To assess Scapholunate ligament instability Procedure • Apply pressure over the scaphoid tubercle • Move the wrist from ulnar deviation to radial deviation Positive Sign Pain or palpable clunk Indicates Scapholunate ligament injury --- **IV THUMB TEST** --- **6 GRIND TEST** Purpose To assess 1st Carpometacarpal CMC joint osteoarthritis Procedure • Apply axial compression to the thumb • Simultaneously rotate the thumb metacarpal Positive Sign Pain or crepitus Indicates 1st CMC joint osteoarthritis --- **I TESTS FOR BONE LIGAMENT CAPSULE JOINT INSTABILITY** --- **1 ULNAR FOVEA TEST** Purpose To assess Triangular Fibrocartilage Complex TFCC injury Procedure • Press the examiner's thumb into the ulnar fovea between the ulnar styloid and flexor carpi ulnaris tendon Positive Sign Pain or tenderness over the ulnar fovea Indicates TFCC injury or ulnotriquetral ligament injury --- **2 ULNAR MENISCO TRIQUETRAL DORSAL GLIDE TEST** Purpose To assess ulnotriquetral ligament and TFCC pathology Procedure • Stabilize the distal ulna • Apply a dorsal glide to the triquetrum while assessing pain and movement Positive Sign Pain or excessive movement Indicates Ulnotriquetral ligament injury or TFCC lesion --- **3 BUNNELL LITTLER TEST** Purpose To differentiate intrinsic muscle tightness from joint capsule tightness Procedure • Flex the PIP joint with the MCP joint extended • Repeat with the MCP joint flexed Positive Sign • PIP flexion improves when MCP is flexed Intrinsic muscle tightness • PIP flexion remains limited Capsular tightness Indicates Intrinsic muscle contracture or capsular restriction // // // // **7 Special Test: Lumbo-Sacral region:** Test for Lumbar Instability: Passive lumbar extension Test etc. Test for Muscle Dysfunction: 90-90 Straight leg rising, Trendelenburg Test etc. Test for Neurological Dysfunction: SLR, Slump Test etc. Test for Joint Dysfunction: Schober's, Faber- Patrick's, Gaenslen, Gillet, March's Test etc. PFDP 7 lumbosacral special tests.pdf **SPECIAL TESTS LUMBO SACRAL REGION** --- **I TEST FOR LUMBAR INSTABILITY** --- **1 PASSIVE LUMBAR EXTENSION TEST PURPOSE** To assess lumbar spine instability PROCEDURE • Patient lies prone • Examiner lifts both the patient's legs together by holding around the ankles • The legs are elevated while the knees remain extended • Observe the patient's symptoms and lumbar response POSITIVE SIGN • Development/reproduction of low back pain • Feeling of heaviness or instability in the lumbar region INDICATES Lumbar segmental instability --- **II TESTS FOR MUSCLE DYSFUNCTION** --- **2 90 90 STRAIGHT LEG RAISING TEST PURPOSE** To assess hamstring muscle tightness/flexibility PROCEDURE • Patient lies supine • Flex the hip and knee to approximately 90 • Stabilize the thigh • Slowly extend the knee while maintaining hip flexion POSITIVE SIGN • Inability to achieve full knee extension • Increased resistance/tightness in the posterior thigh INDICATES Hamstring tightness/shortness --- **3 TRENDLENBURG TEST PURPOSE** To assess hip abductor muscle function especially gluteus medius PROCEDURE • Patient stands on one leg • Ask the patient to lift the opposite foot from the floor • Observe the position of the pelvis POSITIVE SIGN • Pelvis drops on the side of the lifted/non weight bearing leg INDICATES Weakness of the hip abductors of the weight bearing leg particularly gluteus medius --- **III TESTS FOR NEUROLOGICAL DYSFUNCTION** --- **4 STRAIGHT LEG RAISING TEST SLR/LASÈGUES TEST PURPOSE** To assess lumbosacral nerve root irritation particularly due to PIVD PROCEDURE • Patient lies supine • Keep the knee fully extended • Slowly raise the patient's straight leg by flexing the hip • Note the angle and symptoms produced POSITIVE SIGN • Reproduction of radicular/sciatic pain typically between about 30 70 of hip flexion INDICATES Lumbosacral nerve root irritation/compression commonly associated with PIVD --- **5 SLUMP TEST PURPOSE** To assess neural tissue mechanosensitivity particularly lumbosacral nerve root involvement PROCEDURE Patient sits with 1 Hands behind back 2 Thoracic/lumbar spine flexed 3 Cervical spine flexed 4 One knee is extended 5 Ankle is dorsiflexed The examiner

progressively adds these components while observing symptoms **POSITIVE SIGN** • Reproduction of the patients familiar neural/radicular symptoms • Symptoms change with release of one component particularly cervical extension or ankle movement **INDICATES** Neural tissue/nerve root involvement such as lumbar radiculopathy --- **IV TESTS FOR JOINT DYSFUNCTION** --- **6 SCHOBER'S TEST** **PURPOSE** To assess lumbar spine flexion/mobility **PROCEDURE** • Patient stands erect • Mark the lumbosacral region and a point 10 cm above it • Ask the patient to bend forward maximally • Measure the distance between the marks again **POSITIVE SIGN** Insufficient increase in distance during forward flexion **INDICATES** Reduced lumbar flexion/mobility commonly seen in conditions such as ankylosing spondylitis --- **7 FABER / PATRICK'S TEST** **FABER** Flexion ABduction External Rotation **PURPOSE** To assess hip and sacroiliac joint pathology **PROCEDURE** • Patient lies supine • Place the tested leg in • Hip flexion • Abduction • External rotation • Place the lateral ankle over the opposite thigh • Stabilize the opposite pelvis • Gently press the flexed knee downward **POSITIVE SIGN** • Pain reproduced in the hip/groin suggests hip pathology • Pain in the sacroiliac region may indicate SI joint pathology • Restricted ROM may also be significant **INDICATES** Hip or sacroiliac joint dysfunction --- **8 GAENSLER'S TEST** **PURPOSE** To assess sacroiliac joint dysfunction/pathology **PROCEDURE** • Patient lies near the edge of the table • One hip and knee are maximally flexed toward the chest • The opposite leg is allowed to hang off the edge of the table • Apply gentle pressure to produce opposite movements at the pelvis **POSITIVE SIGN** • Reproduction of pain in the sacroiliac region **INDICATES** Sacroiliac joint dysfunction --- **9 GILLET'S TEST** **PURPOSE** To assess sacroiliac joint mobility **PROCEDURE** • Patient stands • Examiner palpates the PSIS and sacrum • Patient flexes one hip and knee lifting the foot from the floor • Observe movement of the PSIS relative to the sacrum **POSITIVE SIGN** • Abnormal or restricted movement of the PSIS compared with the opposite side **INDICATES** Sacroiliac joint hypomobility/dysfunction --- **10 MARCH'S TEST** **PURPOSE** To assess sacroiliac joint movement/dysfunction **PROCEDURE** • Patient stands • Examiner palpates the PSIS • Patient marches by lifting one knee at a time • Observe the movement of the PSIS **POSITIVE SIGN** • Abnormal/asymmetrical movement of the PSIS **INDICATES** Sacroiliac joint dysfunction or altered mobility // 8 **Special Test:** Hip Joint: Test For hip Pathology: Hip scour Test, Leg Roll Test, Patrick's etc. Labral Lesion: Anterior Labral tear Test etc. Pediatric Hip Pathology: Barlow's Test, Galeazzi Sign, Ortolani Sign etc. Test For Muscle dysfunction: Hip Lag sign, 90-90 Straight leg rising, Thomas, Nelaton's line, Bryant's triangle, Faber-Gaenslen, Gillet, March's test, Finkelstein Test, Tripod sign, Trendelenburg sign etc. PFDP 8 hip special tests.pdf --- **I TESTS FOR HIP PATHOLOGY** --- **1 HIP SCOUR TEST** **PURPOSE** To assess for intraarticular hip pathology **PROCEDURE** • Patient lies supine • Flex the hip and knee • Apply a gentle axial load through the femur • Move the hip through flexion/extension and abduction/adduction while maintaining compression **POSITIVE SIGN** • Pain • Catching • Clicking • Crepitus • Restricted/abnormal movement **INDICATES** Possible intraarticular hip pathology such as degenerative or labral pathology -- **2 LEG ROLL TEST** **PURPOSE** To assess for intraarticular hip pathology and restriction of passive hip rotation **PROCEDURE** • Patient lies supine with the leg relaxed • Examiner gently rolls the leg from side to side producing internal and external rotation of the hip • Compare with the opposite side **POSITIVE SIGN** • Pain • Markedly increased or decreased hip rotation • Excessive external rotation compared with the opposite side **INDICATES** Possible hip joint pathology --- **II TEST FOR LABRAL LESION** --- **4 ANTERIOR LABRAL TEAR TEST** **PURPOSE** To assess for an acetabular labral lesion particularly an anterior labral lesion **PROCEDURE** • Patient lies supine • Flex the hip and bring it through the provocative combination of hip movement used to stress the anterior labrum • Reproduce the patients familiar symptoms • Compare with the opposite side **POSITIVE SIGN** • Reproduction of anterior hip/groin pain • Clicking catching or givingway sensation may occur **INDICATES** Possible acetabular labral lesion --- **III PEDIATRIC HIP PATHOLOGY** --- **5 BARLOW'S TEST** **PURPOSE** To determine whether an infants hip is unstable and can be dislocated **PROCEDURE** • Infant lies supine • Flex the hip and knee • Stabilize the pelvis • Gently adduct the hip • Apply gentle posteriorly directed pressure through the femur **POSITIVE SIGN** • A palpable clunk as the femoral head dislocates

INDICATES Dislocatable/unstable hip suggestive of developmental dysplasia of the hip DDH --- **6 GALEAZZI SIGN** **PURPOSE** To detect apparent femoral shortening particularly due to hip dislocation **PROCEDURE** • Infant lies supine • Flex both hips and knees to approximately 90 • Place the feet together • Compare the height of both knees **POSITIVE SIGN** • One knee is lower than the other **INDICATES** • Developmental hip dislocation/dysplasia • Femoral shortening • Other limb length discrepancy --- **7 ORTOLANI SIGN** **PURPOSE** To detect a dislocated hip that can be reduced **PROCEDURE** • Infant lies supine with hips and knees flexed • Stabilize the pelvis • Gently abduct the hip while applying pressure over the greater trochanter in an appropriate direction to reduce the femoral head **POSITIVE SIGN** • A palpable clunk as the femoral head reduces into the acetabulum **INDICATES** Reducible hip dislocation DDH Barlow vs Ortolani Barlow OUT dislocates Ortolani IN reduces -- **IV TESTS FOR MUSCLE DYSFUNCTION** --- **8 HIP LAG SIGN** **PURPOSE** To assess hip extensor muscle function particularly the ability to actively maintain hip extension **PROCEDURE** • Patient lies prone • Examiner passively extends the hip • Ask the patient to maintain the hip in the extended position after the examiner releases support **POSITIVE SIGN** • The patient cannot maintain the extended position • The hip falls back toward flexion **INDICATES** Hip extensor weakness/dysfunction -- **10 THOMAS TEST** **PURPOSE** To assess hip flexor tightness/hip flexion contracture **PROCEDURE** • Patient lies supine near

the edge of the table • Patient flexes one hip and knee toward the chest to flatten the lumbar spine • Observe the position of the opposite leg **POSITIVE SIGN** • Opposite thigh rises from the table hip flexor tightness • Knee extends possible rectus femoris tightness • Hip abducts possible TFL/ITB tightness **INDICATES** Hip flexor muscle tightness/contracture --- **V TESTS FOR HIP DISPLACEMENT** --- **11 NELATONS LINE** **PURPOSE** To assess the position of the greater trochanter in relation to the pelvis **PROCEDURE** • Patient lies in the appropriate position with the hip examined • Draw an imaginary line from the ASIS to the ischial tuberosity • Assess the position of the greater trochanter relative to this line **POSITIVE FINDING** • Greater trochanter lies above Nelatons line **INDICATES** Possible superior displacement of the femur/hip dislocation --- **12 BRYANTS TRIANGLE** **PURPOSE** To detect proximal displacement of the greater trochanter and apparent shortening of the femur **PROCEDURE** • Patient lies supine • From the ASIS draw a perpendicular line to the bed • Draw a line from the greater trochanter to meet this perpendicular • Compare the triangles on both sides **POSITIVE FINDING** • Shortening of the base of Bryants triangle on the affected side **INDICATES** Possible hip dislocation or proximal displacement of the femur --- **VI SACROILIAC JOINT TESTS** --- **14 GAENSLER'S TEST** **PURPOSE** To assess SI joint pathology **PROCEDURE** • Patient lies near the edge of the table • One hip is flexed toward the chest • The opposite leg hangs over the edge • Apply gentle pressure to produce opposite pelvic movements **POSITIVE SIGN** • Reproduction of pain around the SI joint **INDICATES** Sacroiliac joint dysfunction/pathology --- **VII OTHER MUSCLE DYSFUNCTION TESTS** --- **17 TRIPOD SIGN** **PURPOSE** To assess hamstring tightness **PROCEDURE** • Patient sits with the legs extended • Examiner observes the patients sitting posture • Patient attempts to sit upright **POSITIVE SIGN** • Patient leans backward and supports themselves with their hands • Knee flexion may occur because of hamstring tightness **INDICATES** Hamstring tightness/shortness // 9 **Special Test:** Knee Joint: Tests for Ligament Integrity (Collateral & Cruciate): Valgus Stress Test, Varus Stress Test, Lachman Test, Anterior Drawer Test, Posterior Drawer Test etc. Tests for Meniscal Injury: McMurray's Test, Apley's Compression & Distraction Test etc. Tests for Joint Effusion & Patellofemoral Dysfunction: Fluctuation Test, Patellar Tap Test, Clarke's Test etc. Other Measurements: Q-Angle PFDP 9 knee special tests.pdf --- **SPECIAL TESTS KNEE JOINT** --- **I TESTS FOR LIGAMENT INTEGRITY** --- **A COLLATERAL LIGAMENTS** --- **1 VALGUS STRESS TEST** **PURPOSE** To assess the integrity of the medial collateral ligament MCL **PROCEDURE** • Patient lies supine • Knee is slightly flexed approximately 20-30° to unlock the joint • Stabilize the lateral aspect of the knee • Apply a valgus force to the lower leg while stabilizing the thigh **POSITIVE SIGN** • Excessive medial joint opening compared with the opposite side • Pain along the medial knee **INDICATES** MCL injury/insufficiency --- **2 VARUS STRESS TEST** **PURPOSE** To assess the integrity of the lateral collateral ligament LCL **PROCEDURE** • Patient lies supine • Knee is slightly flexed approximately 20-30° • Stabilize the medial aspect of the knee • Apply a varus force to the lower leg **POSITIVE SIGN** • Excessive lateral joint opening • Pain along the lateral knee **INDICATES** LCL injury/insufficiency --- **B CRUCIATE LIGAMENTS** --- **3 LACHMAN TEST** **PURPOSE** To assess the integrity of the anterior cruciate ligament ACL **PROCEDURE** • Patient lies supine • Knee flexed approximately 20-30° • Stabilize the femur with one hand • Hold the proximal tibia with the other hand • Pull the tibia anteriorly **POSITIVE SIGN** • Excessive anterior translation of tibia • Soft/absent firm end feel compared with the normal side **INDICATES** ACL injury/tear Very important Lachman is generally more sensitive for ACL injury than the anterior drawer test particularly in the acute knee --- **4 ANTERIOR DRAWER TEST** **PURPOSE** To assess the ACL **PROCEDURE** • Patient lies supine • Hip flexed and knee flexed to 90° • Stabilize the foot • Grasp the proximal tibia with both hands • Pull the tibia anteriorly **POSITIVE SIGN** • Excessive anterior movement of the tibia compared with the opposite side **INDICATES** ACL injury/insufficiency --- **5 POSTERIOR DRAWER TEST** **PURPOSE** To assess the integrity of the posterior cruciate ligament PCL **PROCEDURE** • Patient lies supine • Knee flexed to approximately 90° • Stabilize the foot • Place both hands around the proximal tibia • Push the tibia posteriorly **POSITIVE SIGN** • Excessive posterior translation of the tibia **INDICATES** PCL injury/insufficiency --- **II TESTS FOR MENISCAL INJURY** --- **6 MCMURRAY'S**

TEST **PURPOSE** To assess for medial and lateral meniscal lesions **PROCEDURE** • Patient lies supine • Fully flex the knee • For medial meniscus externally rotate the tibia and apply valgus stress while extending the knee • For lateral meniscus internally rotate the tibia and apply varus stress while extending the knee • Palpate the joint line **POSITIVE SIGN** • Pain • Palpable/audible click or clunk at the joint line **INDICATES** Possible meniscal tear --- **7 APLEY'S COMPRESSION DISTRACTION TEST** **PURPOSE** To differentiate meniscal pathology from ligament pathology **PROCEDURE** Compression • Patient lies prone • Knee flexed to 90° • Stabilize the thigh • Apply downward compression through the heel • Rotate the tibia internally and externally Distraction • Maintain the same position • Apply traction to the tibia while rotating it **POSITIVE SIGN** Compression positive • Pain reproduced with compression/rotation Suggests meniscal pathology Distraction positive • Pain reproduced with traction/rotation Suggests ligament/capsular pathology --- **III JOINT EFFUSION PATELOFEMORAL DYSFUNCTION** --- **8 FLUCTUATION TEST** **PURPOSE** To detect knee joint effusion particularly larger effusions **PROCEDURE** • Patient lies supine with the knee relaxed/extended • Examiner compresses the fluid from one side of the knee toward the opposite side • Observe/palpate for a fluid wave **POSITIVE SIGN** • A palpable/visible fluid wave or fluctuation **INDICATES** Knee joint effusion --- **9 PATELLAR TAP TEST** **PURPOSE** To detect knee joint effusion **PROCEDURE** • Patient lies supine with knee extended • Gently press/compress the suprapatellar region to move fluid beneath the patella • Tap the patella

downward against the femur **POSITIVE SIGN** • Patella rebounds/floats and taps the femur producing a palpable tap **INDICATES** Significant knee joint effusion --- **10 CLARKES TEST / PATELLAR GRIND TEST PURPOSE** To assess patellofemoral dysfunction particularly chondromalacia/patellofemoral pain **PROCEDURE** • Patient lies supine with the knee extended • Examiner places the hand proximal to the patella • Apply gentle downward pressure to the patella • Ask the patient to contract the quadriceps **POSITIVE SIGN** • Pain behind/around the patella • Inability to maintain quadriceps contraction because of pain **INDICATES** Possible patellofemoral dysfunction/chondral pathology --- **IV OTHER MEASUREMENT --- 11 Q ANGLE PURPOSE** To assess patellofemoral alignment and the lateral pull on the patella **MEASUREMENT** Q angle is formed by the intersection of • Line 1 ASIS centre of patella • Line 2 centre of patella tibial tubercle **PROCEDURE** • Patient positioned with the knee in the appropriate standardized position • Identify and mark • ASIS • Centre of patella • Tibial tubercle • Use a goniometer to measure the angle between the two lines **POSITIVE/ABNORMAL FINDING** An increased Q angle may indicate increased lateral patellar alignment/pull and may be associated with patellofemoral dysfunction **INDICATES** • Patellofemoral malalignment • Predisposition to lateral patellar tracking/instability **10 Special Test: Ankle & Foot Complex: Tests for Ligament Stability: Anterior Drawer Test, Talar Tilt Test etc. Tests for Vascular Conditions (e.g., DVT): Homan's Test, Moses Test etc. PFDP 10 ankle foot special tests.pdf --- I TESTS FOR LIGAMENT STABILITY --- 1 ANTERIOR DRAWER TEST ANKLE PURPOSE** To assess the integrity of the anterior talofibular ligament ATFL and lateral ankle stability **PROCEDURE** • Patient sits or lies supine • Keep the ankle in a neutral or slight plantar flexed position • Stabilize the distal tibia with one hand • Grasp the heel/talus with the other hand • Draw the foot/talus anteriorly **POSITIVE SIGN** • Excessive anterior translation compared with the opposite ankle • Soft/absent end feel • Pain may be reproduced **INDICATES** ATFL injury/lateral ankle instability --- **2 TALAR TILT TEST PURPOSE** To assess the integrity of the lateral ankle ligaments particularly the calcaneofibular ligament CFL **PROCEDURE** • Patient sits or lies supine • Stabilize the distal leg • Grasp the calcaneus/heel • Tilt the calcaneus into inversion **POSITIVE SIGN** • Excessive inversion compared with the opposite side • Pain or a soft end feel **INDICATES** CFL injury/lateral ankle instability --- **II TESTS FOR VASCULAR CONDITIONS DVT --- 3 HOMANS TEST PURPOSE** Traditionally used to assess for deep vein thrombosis DVT of the lower limb **PROCEDURE** • Patient lies supine • Knee is extended • Examiner supports the leg • Passively dorsiflex the ankle **POSITIVE SIGN** • Pain in the calf during passive ankle dorsiflexion **INDICATES** Traditionally considered suggestive of DVT --- **4 MOSES TEST PURPOSE** Traditionally used to assess for DVT **PROCEDURE** • Patient lies supine with the leg relaxed • Compress the calf anteroposteriorly • Then compress the calf laterally **POSITIVE SIGN** • Pain produced by anteroposterior compression but not by lateral compression **INDICATES** Traditionally considered suggestive of DVT **SPECIAL TESTS --- I TESTS FOR CIRCULATION and SWELLING 1 ALLEN TEST Purpose** To assess the patency of the radial and ulnar arteries **Procedure** • Patient clenches the fist Compress both radial and ulnar arteries Patient opens the hand Release one artery at a time and observe colour return **Positive Sign** Delayed or absent return of normal hand colour **Indicates** Arterial insufficiency or vascular occlusion **2 DIGITAL BLOOD FLOW TEST Purpose** To assess circulation to the fingers **Procedure** • • Compress the fingertip until it blanches Release the pressure **Positive Sign** Delayed capillary refill greater than 2 seconds **Indicates** Reduced peripheral circulation **3 FIGURE OF 8 MEASUREMENT Purpose** To assess hand and wrist swelling edema **Procedure** • Use a measuring tape in a standardized figure of 8 pattern around the wrist and hand Record the measurement **1 • Positive Finding** Increased measurement compared with the opposite side **Indicates** Hand or wrist edema **4 HAND VOLUME TEST Purpose** To assess hand edema **Procedure** • • Immerse the hand into a volumeter filled with water Measure the volume of displaced water **Positive Finding** Increased displaced volume **Indicates** Hand swelling/edema --- **II TESTS FOR THORACIC OUTLET SYNDROME 1 ADSONS TEST Purpose** To assess Thoracic Outlet Syndrome **TOS Procedure** • • Palpate the radial pulse Patient extends the neck rotates the head toward the tested side and takes a deep breath **Positive Sign** Reduced/absent radial pulse or reproduction of symptoms **Indicates** Thoracic Outlet Syndrome **2 ROOS STRESS TEST EAST Purpose** To assess Thoracic Outlet Syndrome **Procedure** Patient abducts shoulders to 90 degrees externally rotates them and flexes elbows to 90 degrees **2 • Patient repeatedly opens and closes the hands for 3 minutes Positive Sign** Pain numbness tingling heaviness or inability to continue **Indicates** Thoracic Outlet Syndrome

--- **1 Blood Investigations and Cardiac Biomarkers • Definition** • Blood investigations and biomarkers are laboratory tests performed on venous or arterial blood samples to evaluate organ function systemic inflammation hematological balance and myocardial cellular damage or heart failure • **Key Cardiac and Hematological Markers** • Cardiac Troponins cTnI and cTnT • Gold standard regulatory contractile proteins released exclusively upon cardiomyocyte necrosis • Normal Value cTnI is less than 0 point 04 ng/mL high sensitivity troponin hs cTnI is less than 14 ng/L • Timeline Rises within 3 to 4 hours post myocardial injury peaks at 18 to 24 hours and remains elevated for 10 to 14 days • Clinical Significance Confirms Acute Myocardial Infarction AMI determines the extent of myocardial necrosis • Creatine Kinase MB CK MB • Isoenzyme localized primarily in cardiac muscle tissue • Normal Value Less than 5 percent of total CK or 0 to 5 ng/mL • Timeline Rises within 4 to 6 hours peaks at 18 to 24 hours and normalizes within 48 to 72 hours • Clinical Significance Useful for detecting early re infarction because it clears much faster than troponin • B Type Natriuretic Peptide BNP and NT proBNP • Neurohormone

secreted by ventricular myocytes in response to increased ventricular wall stretch and volume overload • Normal Value BNP is less than 100 pg/mL NT proBNP is less than 300 pg/mL • Clinical Significance Distinguishes dyspnea caused by congestive heart failure elevated levels from dyspnea of primary pulmonary origin • Complete Blood Count CBC and Hemoglobin Hb • Normal Hemoglobin 13 to 17 g/dL for males 12 to 15 g/dL for females • Clinical Significance Anemia Hb less than 8 g/dL drastically impairs peripheral oxygen delivery causing resting tachycardia fatigue and reduced exercise tolerance • Inflammatory Markers CRP and ESR • High Sensitivity C Reactive Protein hs CRP Values greater than 3 point 0 mg/L reflect systemic inflammation and elevated atherothrombotic risk • Erythrocyte Sedimentation Rate ESR Elevated in pericarditis infective endocarditis and chronic pulmonary infections • Coagulation Profile D Dimer PT INR • D Dimer Normal is less than 500 ng/mL elevated levels indicate active fibrinolysis and raise suspicion for deep vein thrombosis DVT or pulmonary embolism PE • International Normalized Ratio INR Normal is 0 point 8 to 1 point 2 therapeutic range for anticoagulated patients is 2 point 0 to 3 point 0 • **Physiotherapy Implications and Exercise Precautions** • Active myocardial infarction with progressively rising troponin levels is an absolute contraindication to physical therapy exercise • Ambulation or phase I cardiac rehabilitation is deferred until cardiac enzymes peak and decline and the patient remains hemodynamically stable for 24 hours • If Hemoglobin is less than 8 g/dL postpone active resistance training limit interventions to gentle bedside bed mobility or assisted transfers • If INR is greater than 3 point 5 avoid vigorous chest physiotherapy manual percussions aggressive stretching or high impact mobilizations due to internal bleeding and hematoma risk • Severely elevated BNP indicates decompensated heart failure fluid overload requires avoiding head down postural drainage and minimizing cardiovascular exertion --- **2 Arterial Blood Gas ABG Analysis • Definition** • ABG analysis is an invasive diagnostic measurement of arterial blood assessing systemic oxygenation alveolar ventilation and metabolic or respiratory acid base balance • **Standard Arterial Parameters and Reference Ranges** • pH 7 point 35 to 7 point 45 indicates systemic hydrogen ion concentration • PaCO₂ Partial pressure of arterial carbon dioxide 35 to 45 mmHg respiratory/ventilatory component • PaO₂ Partial pressure of arterial oxygen 80 to 100 mmHg on room air arterial oxygenation status • HCO₃ Bicarbonate concentration 22 to 26 mEq/L metabolic/renal component • Base Excess BE minus 2 to plus 2 mEq/L amount of buffer required to restore normal pH • SaO₂ Arterial oxygen saturation 95 percent to 100 percent • **Systematic Steps for ABG Interpretation** • Step 1 Check pH A pH less than 7 point 35 indicates acidemia a pH greater than 7 point 45 indicates alkalemia • Step 2 Check PaCO₂ If altered in the direction opposite to pH low pH with high PaCO₂ the primary disorder is respiratory • Step 3 Check HCO₃ If altered in the same direction as pH low pH with low HCO₃ the primary disorder is metabolic • Step 4 Determine Compensation • Uncompensated Abnormal pH one component abnormal other component normal • Partially Compensated Abnormal pH both PaCO₂ and HCO₃ abnormal in the same compensatory direction • Fully Compensated Normal pH within 7 point 35 to 7 point 45 but both PaCO₂ and HCO₃ remain abnormal • **Common Pathological Patterns** • Respiratory Acidosis pH less than 7 point 35 PaCO₂ greater than 45 mmHg • Causes Alveolar hypoventilation severe COPD respiratory muscle fatigue central nervous system depression Guillain Barre syndrome • Respiratory Alkalosis pH greater than 7 point 45 PaCO₂ less than 35 mmHg • Causes Hyperventilation secondary to anxiety hypoxemia acute pulmonary embolism severe pain • Metabolic Acidosis pH less than 7 point 35 HCO₃ less than 22 mEq/L • Causes Diabetic ketoacidosis lactic acidosis from septic or cardiogenic shock chronic renal failure • Metabolic Alkalosis pH greater than 7 point 45 HCO₃ greater than 26 mEq/L • Causes Prolonged vomiting high dose diuretic administration severe hypokalemia • **Physiotherapy Significance** • Identifies type I respiratory failure hypoxemic PaO₂ less than 60 mmHg with normal/low PaCO₂ vs type II respiratory failure hypercapnic PaCO₂ greater than 50 mmHg with acidosis • Guides non invasive ventilation BiPAP and aggressive airway clearance techniques for hypercapnic retained secretions in COPD • Prevents over oxygenation in retainers uncontrolled high flow oxygen suppresses the hypoxic drive to breathe in chronic type II respiratory failure • Serves as an objective parameter for monitoring tolerance during ventilator weaning and progressive bedside mobilization --- **3 Pulmonary Function Testing PFT • Definition** • Pulmonary Function Testing encompasses non invasive physiological evaluations primarily spirometry plethysmography and gas diffusion capacity used to quantify static and dynamic lung volumes airway mechanics and gas exchange efficiency • **Key Spirometric Parameters** • Tidal Volume TV Normal volume inspired or expired during quiet breathing approximately 500 mL • Vital Capacity VC / Forced Vital Capacity FVC Maximum volume of air exhaled forcefully following maximal inspiration normal is greater than 80 percent predicted • Forced Expiratory Volume in 1 Second FEV₁ Volume exhaled during the first second of an FVC maneuver normal is greater than 80 percent predicted • FEV₁/FVC Ratio Proportion of total vital capacity cleared in the first second normal reference is 70 percent to 85 percent 0 point 70 to 0 point 85 • Peak Expiratory Flow Rate PEFR Maximum flow velocity reached during forced expiration monitored in asthma management • Diffusing Capacity of the Lungs for Carbon Monoxide DLCO Assesses alveolar capillary membrane gas transfer normal is greater than 75 percent predicted • **Diagnostic Classification of Patterns** • Obstructive Lung Pattern • Characterized by expiratory flow limitation and increased airway resistance • Criteria FEV₁/FVC ratio is less than 0 point 70 70 percent FEV₁ is reduced FVC may be normal or reduced Total Lung Capacity TLC and Residual Volume RV are elevated due to air trapping and hyperinflation •

Examples COPD Chronic Bronchitis Emphysema Bronchial Asthma Bronchiectasis Cystic Fibrosis • Reversibility Testing An improvement in FEV1 by greater than 12 percent and at least 200 mL after inhaling a short acting bronchodilator indicates reversible airway obstruction classic for Asthma • Restrictive Lung Pattern • Characterized by reduced lung compliance restricted expansion and reduced lung volumes • Criteria FEV1/FVC ratio is normal or elevated greater than 0 point 70 FVC is less than 80 percent predicted TLC is less than 80 percent predicted • Intrinsic Causes Idiopathic Pulmonary Fibrosis Sarcoidosis Asbestosis ARDS • Extrinsic Causes Kyphoscoliosis ankylosing spondylitis diaphragmatic paralysis obesity hypoventilation syndrome • Diffusion Impairment • Reduced DLCO with restrictive mechanics points to alveolar wall thickening eg Idiopathic Pulmonary Fibrosis • Reduced DLCO with obstructive mechanics points to alveolar wall destruction eg Emphysema • **Relevance in Physiotherapy Management** • Differentiates whether breathing exercises should focus on relieving flow obstruction or expanding lung volumes • In obstructive disease Emphasize prolonged expiratory phases pursed lip breathing pacing techniques and active cycle of breathing techniques ACBT • In restrictive disease Focus on inspiratory capacity enhancement thoracic expansion exercises incentive spirometry and rib cage mobilization • Establishes baseline aerobic exercise tolerance and monitors pulmonary rehabilitation outcomes --- **4 Electrocardiography**

ECG • Definition • An electrocardiogram is a surface recording of the electrical potentials generated by cardiac muscle depolarization and repolarization over time recorded via a standard 12 lead electrode configuration • **Normal ECG**

Deflections Intervals and Values • P Wave Atrial depolarization normal duration is less than 0 point 12 seconds 3 small boxes amplitude is less than 2 point 5 mm • PR Interval Conduction travel time from the sinoatrial node through the AV node to ventricles normal duration is 0 point 12 to 0 point 20 seconds 3 to 5 small boxes • QRS Complex Ventricular depolarization normal duration is 0 point 06 to 0 point 10 seconds less than 3 small boxes • ST Segment Interval between ventricular depolarization and repolarization normally sits flat on the isoelectric baseline • T Wave Ventricular repolarization smooth asymmetrical deflection oriented in the same general direction as the main QRS deflection • QT Interval Total time for ventricular activation and recovery normal corrected QT QTc is less than 0 point 44 seconds

• **Common Pathological Conditions** • Myocardial Ischemia Injury and Infarction • Ischemia Horizontal or downsloping ST segment depression greater than or equal to 1 mm below baseline or symmetrical T wave inversion • Acute Injury ST segment elevation STEMI greater than 1 mm in limb leads or greater than 2 mm in precordial leads measured at the J point • Established/Old Infarction Pathological Q waves duration greater than 0 point 04 seconds or depth greater than 25 percent of the corresponding R wave amplitude • Cardiac Arrhythmias • Atrial Fibrillation AF Absent distinct P waves irregular chaotic baseline fibrillatory f waves irregularly irregular ventricular rhythm • Premature Ventricular Contractions PVCs Early wide bizarre QRS complexes greater than 0 point 12 seconds without preceding P waves followed by a full compensatory pause • Ventricular Tachycardia VT Run of three or more consecutive wide QRS complexes at a rate greater than 100 beats per minute medical emergency • Conduction Blocks • First Degree AV Block Fixed PR interval prolongation greater than 0 point 20 seconds with every P wave followed by a QRS • Second Degree AV Block Mobitz I / Wenckebach Progressive PR prolongation until a QRS complex is dropped • Second Degree AV Block Mobitz II Constant PR interval with intermittently dropped QRS complexes • Third Degree Complete AV Block Complete AV dissociation P waves and wide escape QRS complexes fire at independent intrinsic rates • **Physiotherapy Termination Criteria During Exercise** • Exercise must be terminated immediately if any of the following occur on ECG telemetry • ST segment depression or elevation greater than 1 to 2 mm • Sustained runs of ventricular tachycardia or multifocal PVCs • Appearance of new atrial fibrillation or progressive AV block • Sudden drop in systolic blood pressure accompanied by ischemic ECG patterns --- **5**

Echocardiography • Definition • Echocardiography is a non invasive diagnostic ultrasound imaging modality that evaluates cardiac anatomy chamber dimensions myocardial wall motion valvular structure and hemodynamic blood flow patterns

• **Common Modalities** • Transthoracic Echocardiography TTE Standard non invasive scan through anterior intercostal acoustic windows • Transesophageal Echocardiography TEE Endoscopic probe passed into the esophagus provides high resolution imaging of posterior structures atrial appendages and prosthetic valves • Doppler Echocardiography Quantifies blood flow velocities pressure gradients across valves and regurgitant flow volumes • Stress Echocardiography Compares resting myocardial wall motion with post exercise or pharmacologically induced stress using Dobutamine to identify inducible ischemia • **Key Hemodynamic and Structural Parameters** • Left Ventricular Ejection Fraction LVEF • Percentage of end diastolic volume ejected with each systolic contraction • Normal 55 percent to 70 percent • Mild Dysfunction 40 percent to 54 percent • Moderate Dysfunction 30 percent to 39 percent • Severe Dysfunction Less than 30 percent • Regional Wall Motion Abnormalities RWMA • Evaluates segmental myocardial contractility hypokinesia reduced movement akinesia no movement or dyskinesia paradoxical outward bulging in systole confirms ischemic territories • Valvular Pathology • Stenosis valve orifice narrowing high pressure gradients vs Regurgitation valve leaflets fail to coapt causing backward regurgitant jet • Diastolic Function Parameters • E/A ratio ratio of early mitral inflow velocity to late atrial filling velocity abnormal filling pressures indicate heart failure with preserved ejection fraction HFpEF • **Relevance in Physiotherapy and Cardiac Rehabilitation** • Safe exercise prescription relies directly on baseline LVEF • Patients with LVEF

less than 35 percent require continuous telemetry longer warm up and cool down phases and strict low to moderate initial workloads RPE 11 to 13 on the Borg 6 to 20 scale • Severe symptomatic aortic stenosis peak aortic velocity greater than 4 m/s or valve area less than 1 point 0 cm² is an absolute contraindication to vigorous exercise training due to syncope and sudden death risks • Echo derived pulmonary artery systolic pressure PASP greater than 35 mmHg identifies pulmonary hypertension necessitating avoidance of heavy isometric exercises or Valsalva maneuvers that spike right ventricular afterload ---

6 Coronary Angiography Cardiac Catheterization • Definition • Coronary angiography is an invasive fluoroscopic imaging procedure wherein radiopaque iodinated contrast material is selectively injected into coronary arteries via arterial catheterization to visualize lumen patency and stenosis • **Procedure and Access Sites •** Transfemoral Approach Catheter introduced through the femoral artery in the groin requires extended post procedure supine immobilization • Transradial Approach Catheter introduced through the radial artery at the wrist lower bleeding risk and allows early post procedure ambulation • Contrast is injected under continuous X ray fluoroscopy Cineangiography into the Left Main Coronary Artery Left Anterior Descending LAD Circumflex LCx and Right Coronary Artery RCA • **Diagnostic Findings and Interpretation •** Significant Coronary Stenosis Luminal diameter reduction greater than or equal to 70 percent is considered hemodynamically significant a reduction greater than or equal to 50 percent in the Left Main stem is critical • Single Double or Triple Vessel Disease Identifies the number of main vessels obstructed determines eligibility for Percutaneous Coronary Intervention PCI with stenting vs Coronary Artery Bypass Grafting CABG • Fractional Flow Reserve FFR Guidewire sensor measuring pressure drop across a lesion FFR less than 0 point 80 proves the stenosis causes myocardial ischemia • **Post Procedure Complications •** Retroperitoneal or local entry site hematoma pseudoaneurysm acute arterial occlusion contrast induced nephropathy arrhythmia or catheter induced thromboembolic stroke

• **Physiotherapy Precautions and Mobilization Protocols •** Transfemoral Route Strict bed rest in supine position with the affected leg immobilized and hip kept neutral no flexion greater than 30 degrees for 4 to 6 hours after manual compression or sheath removal • Transradial Route Patient may ambulate within 1 to 2 hours avoid wrist flexion weight bearing through the hand and lifting objects over 2 to 3 kg for 48 to 72 hours • Inspect the puncture site for active bleeding expanding swelling/hematoma or distal pulse loss dorsalis pedis/radial artery before initiating mobilization ---

7 Chest X Ray CXR • Definition • A chest radiograph is a non invasive projection transmission image displaying the differential radiodensity of thoracic tissues air black/radiolucent fat dark gray water/soft tissue light gray and bone/metal white/radiopaque • **Standard Projections •** Posteroanterior PA View Gold standard taken in full upright inspiration with X ray beam travelling from back to front minimizes cardiac magnification • Anteroposterior AP View Performed at bedside on immobilized or ICU patients magnifies cardiac silhouette size and limits scapular clearance • **Systematic Evaluation Approach ABCDE Method •** A Airway Trachea position normally midline vertically oriented over spinous processes • B Breathing / Lung Fields Vascular markings lucency symmetric expansion pleural spaces • C Cardiac Silhouette and Mediastinum Cardiothoracic Ratio CTR on PA view is normally less than 0 point 50 50 percent mediastinum shape and borders • D Diaphragm Both hemidiaphragms rounded smooth right sitting 1 to 2 cm higher than left costophrenic and cardiophrenic angles sharp and clear • E Everything Else / Skeleton Ribs clavicles humeri subcutaneous tissues indwelling tubes catheters pacing wires • **Common Pathological Variations on CXR •** Lobular Consolidation Pneumonia • Homogeneous radio opacity with preservation of lung volume • Presence of Air Bronchograms branching lucent tubular airways outlined by fluid filled alveoli • Silhouette sign Loss of the right heart border indicates right middle lobe consolidation • Pleural Effusion • Blunting of the costophrenic angle evident when fluid exceeds 200 mL • Meniscus sign Homogeneous dense opacity with a concave superior border sloping upward toward the lateral chest wall • Massive effusion displaces the mediastinum and trachea away from the affected side • Pneumothorax • Complete absence of lung markings at the peripheral thoracic cavity • Visible sharp white visceral pleural line separated from the chest wall by radiolucent air • Tension Pneumothorax Marked tracheal and mediastinal shift toward the contralateral side with flattened/depressed ipsilateral hemidiaphragm • Atelectasis Lung Collapse • Opacity accompanied by direct signs of volume loss tracheal cardiac and mediastinal shift toward the side of collapse • Crowding of ribs and elevation of the ipsilateral hemidiaphragm • COPD / Emphysema • Bilateral pulmonary hyperinflation with increased lung field radiolucency • Flattened depressed hemidiaphragms with widened intercostal spaces • Elongated narrow tubular heart shadow enlarged retrosternal clear space on lateral view • Pulmonary Edema Congestive Heart Failure • Cardiomegaly CTR greater than 0 point 50 • Cephalization of pulmonary vessels upper lobe blood diversion • Perihilar Bat wing or Butterfly alveolar infiltrates • Kerley B lines short horizontal peripheral lines representing thickened interlobular septa • **Physiotherapy Implications •** Localizes pulmonary pathologies to specific segments directly guiding postural drainage positioning • Consolidation/Atelectasis confirms the need for thoracic expansion techniques chest percussion vibration and directed autogenic drainage • Pneumothorax is a strict contraindication to positive pressure techniques PEP IPPB and vigorous manual chest clapping until an intercostal chest tube ICD is placed and underwater seal bubbling stabilizes ---

8 Tests for Peripheral Arterial and Venous Conditions • Definition • Clinical and non invasive diagnostic tests evaluating arterial flow

velocity lumen patency tissue perfusion and venous valvular competence in the lower and upper extremities • **Peripheral Arterial Disease PAD Tests** • Ankle Brachial Index ABI • Definition Ratio of the highest systolic blood pressure at the ankle to the highest systolic blood pressure in the brachial arteries • Procedure Patient rests supine for 10 minutes A Doppler ultrasound probe 5 to 10 MHz and pneumatic cuff measure systolic blood pressure in both brachial arteries and bilateral ankle arteries Dorsalis Pedis and Posterior Tibial • Calculation ABI equals Highest Ankle Systolic Pressure DP or PT / Highest Brachial Systolic Pressure Right or Left • Clinical Values and Interpretation • 1 point 00 to 1 point 40 Normal arterial perfusion • 0 point 91 to 0 point 99 Borderline arterial disease • 0 point 41 to 0 point 90 Mild to moderate PAD characteristic intermittent claudication during walking • Less than or equal to 0 point 40 Severe PAD critical limb ischemia rest pain gangrene risk • Greater than 1 point 40 Non compressible calcified rigid arteries commonly seen in diabetes mellitus requires Toe Brachial Index • Buerger's Test Postural Color Changes • The patient lies supine the examiner elevates the lower extremity to 45 to 60 degrees for 1 to 2 minutes • Abnormal Development of marked pallor in less than 60 seconds positive vascular angle of Buerger less than 20 degrees indicates severe ischemia • On returning the leg to a dependent position a delay in pink reperfusion greater than 15 seconds followed by a dark purplish red discoloration Rubor of Dependency confirms chronic arterial insufficiency • Capillary Refill Time CRT • Firm pressure applied to the distal digital pulp/nail bed for 5 seconds until blanched • Normal Color returns in less than 2 seconds prolonged CRT indicates distal hypoperfusion • **Peripheral Venous Disease Tests** • Wells Clinical Score for Deep Vein Thrombosis DVT • Stratifies pre test probability of DVT based on risk factors active cancer plus 1 paralysis/recent plaster immobilization plus 1 bedridden over 3 days or major surgery within 12 weeks plus 1 localized tenderness along deep venous system plus 1 entire leg swollen plus 1 calf swelling greater than 3 cm compared to asymptomatic side plus 1 pitting edema confined to symptomatic leg plus 1 collateral superficial non varicose veins plus 1 alternative diagnosis as likely as DVT minus 2 • Score greater than or equal to 2 DVT likely urgent Duplex Ultrasound required • Homans Sign • Passive dorsiflexion of the ankle with the knee extended to elicit calf pain • Clinical Note Insensitive non specific and carries a mechanical risk of dislodging an unstable clot It is no longer recommended in modern clinical guidelines • Trendelenburg Test Brodie Trendelenburg Test • Differentiates saphenofemoral junction incompetence from deep/perforator valvular incompetence in varicose veins • Leg is elevated to empty superficial veins a tourniquet is applied around the upper thigh • Patient stands rapid retrograde filling from above before releasing the tourniquet indicates perforator incompetence rapid filling from above after tourniquet release confirms saphenofemoral junction valve incompetence • Venous Duplex Ultrasound • Gold standard non invasive diagnostic investigation for DVT evaluates venous compressibility and venous flow patterns • **Physiotherapy Implications and Precautions** • In Intermittent Claudication PAD with ABI 0 point 5 to 0 point 9 Supervised treadmill walking is the primary conservative therapy Exercise to moderate to strong claudication pain grade 3 on 4 point scale rest until pain resolves and repeat for 30 to 45 minutes 3 to 5 times weekly • In Severe PAD ABI less than 0 point 5 High intensity exercise leg elevation and external compression bandages are strictly contraindicated • In Acute Confirmed DVT Hold all ambulation range of motion exercises and calf massage until adequate therapeutic anticoagulation is established typically 24 to 48 hours with Low Molecular Weight Heparin • In Chronic Venous Insufficiency Promote leg elevation above heart level prescribe graduated compression stockings 20 to 30 mmHg and prescribe active calf muscle pump exercises //////////////

Sports Specific Assessment BPT 3rd Year 5 Mark University Exam Answers • **Definition** Sports specific assessment is the systematic evaluation of an athletes physical fitness functional abilities and performance related components required for a particular sport • **Purpose** • To assess fitness and sports performance • To identify strengths and weaknesses • To detect physical limitations and injury risk factors • To monitor training progress and recovery • To help plan individualized sports training and rehabilitation --- **1 Cardiovascular Endurance** • **Definition** Ability of the cardiovascular and respiratory systems to supply oxygen to working muscles during prolonged exercise • **Purpose** To assess aerobic fitness and endurance capacity • **Assessment tests** • Cooper 12 minute run test • 20 metre multistage shuttle run test Beep test • 6 minute walk test • **Parameters** Distance covered heart rate respiratory rate and recovery time • **Example** Beep test for football players --- **2 Muscular Strength** • **Definition** Ability of a muscle or muscle group to produce maximum force against resistance • **Purpose** To assess muscle force required for sports activities • **Assessment tests** • One repetition maximum 1RM test • Handgrip dynamometer test • Isometric strength testing • **Parameters** Maximum force produced and side to side differences • **Example** 1RM squat test for weightlifters --- **3 Flexibility** • **Definition** Ability of a joint to move through its available range of motion • **Purpose** To assess joint mobility and muscle flexibility required for sports movements • **Assessment tests** • Sit and reach test • Goniometric measurement • Straight leg raise test • **Parameters** Range of motion and muscle length • **Example** Sit and reach test for gymnasts --- **4 Speed** • **Definition** Ability to perform a movement or cover a distance in the shortest possible time • **Purpose** To assess rapid movement and sprinting ability • **Assessment tests** • 30 metre sprint test • 40 metre sprint test • 10 metre sprint test • **Parameters** Time taken to complete the distance • **Example** 30 metre sprint test for sprinters and football players --- **5 Agility** • **Definition** Ability to change body direction rapidly and efficiently while maintaining balance and control • **Purpose** To assess quick directional changes during sports

activities • **Assessment tests** • T test • Illinois agility test • 505 agility test • **Parameters** Time taken movement control and directional changes • **Example** T test for basketball players --- **6 Power** • **Definition** Ability to produce maximum force in minimum time Power is the product of force and velocity • **Purpose** To assess explosive muscular performance

• **Assessment tests** • Vertical jump test • Standing broad jump test • Medicine ball throw test • **Parameters** Jump height distance or throwing distance • **Example** Vertical jump test for volleyball players --- **7 Balance** • **Definition** Ability to maintain the body's centre of mass over its base of support • **Types** Static balance and dynamic balance • **Purpose** To assess postural control and stability during sports activities • **Assessment tests** • Single leg stance test • Y Balance test • Star Excursion Balance Test SEBT • **Parameters** Time maintained reach distance and postural stability • **Example** Y Balance test for football players --- **8 Coordination** • **Definition** Ability to perform smooth accurate and controlled movements using different body parts • **Purpose** To assess neuromuscular control and the integration of movements • **Assessment tests** • Alternate hand wall toss test • Finger to nose test • Alternate hand foot tapping test • **Parameters** Accuracy timing rhythm and movement smoothness • **Example** Wall toss test for tennis players --- **9 Reaction Time** • **Definition** Time taken to respond to a stimulus after it is presented • **Types** Simple reaction time and choice reaction time • **Purpose** To assess the speed of response to visual or auditory stimuli • **Assessment tests** • Ruler drop test • Computerized reaction time test • Sport specific stimulus response test • **Parameters** Time taken to respond usually measured in milliseconds • **Example** Reaction time test for a sprinter responding to the starting signal --- **References** • ACSMs Guidelines for Exercise Testing and Prescription • Physiology of Sport and Exercise Kenney Wilmore and Costill

--- **Types of Pain** • **Somatic Pain** • Arises from noxious stimulation of musculoskeletal cutaneous and deep somatic structures • Superficial Somatic Originates in skin or subcutaneous tissues sharp pricking well localized with clearly defined boundaries • Deep Somatic Originates in muscles tendons ligaments periosteum and joint capsules dull aching poorly localized and tends to refer to related sclerotomal structures • **Somatic Referred Pain** • Pain perceived at a site distant from the actual primary site of somatic injury or dysfunction eg cervical facet joint dysfunction referring into the shoulder girdle • Follows an embryological segmental or sclerotomal pattern without mechanical irritation or compression of neural tissue • Characterized by a diffuse deep ache with no primary neurological signs normal reflexes sensation and motor power • **Neurogenic Neuropathic Pain** • Initiated or caused by a primary lesion dysfunction or mechanical irritation of the peripheral or central nervous system • Described as sharp shooting lancinating burning stabbing or electric shock like sensations • Follows discrete anatomical distributions dermatomal tracts or peripheral nerve distributions and frequently presents with positive or negative neurological symptoms paresthesia hyperalgesia allodynia hypesthesia motor deficits • **Visceral Pain** • Arises from the distension ischemia spasm or inflammation of internal organs thoracic abdominal or pelvic • Poorly localized deep sickening dull and cramp like • Frequently accompanied by autonomic manifestations nausea diaphoresis pallor blood pressure changes and visceral somatic referral patterns eg myocardial ischemia referring to the jaw/left arm gallbladder referral to the right inferior scapular region --- **Subjective Assessment of Pain** • **Location** • Pinpointing the exact anatomical site and boundary using a standardized body chart distinguishing localized pain from widespread or multi site pain • **Duration** • Timeframe since the onset of symptoms • Acute Less than 6 weeks direct relation to tissue injury and healing • Subacute 6 to 12 weeks • Chronic / Persistent Greater than 3 to 6 months outlasting normal physiological healing time • **Progression** • The chronological course of the pain over time improving worsening static intermittent episodic or recurrent • **Distribution** • Spatial pattern of symptoms focal segmental/sclerotomal radiating along a dermatomal/peripheral nerve track or non anatomical/diffuse • **Quality** • Subjective descriptive character of the sensation eg sharp dull ache burning throbbing heavy stabbing indicating the underlying tissue source and neurophysiological mechanism • **Diurnal Variations** • Behavior of symptoms across a 24 hour day night cycle • Morning Stiffness and pain lasting more than 30 to 60 minutes points to inflammatory etiology eg rheumatoid arthritis • Evening / End of day Pain aggravated by weight bearing and activity points to mechanical or degenerative etiology eg osteoarthritis • Night Constant unremitting pain waking the patient from sound sleep is a clinical red flag requiring evaluation for non mechanical pathology eg malignancy infection • **Modifying Factors** • Aggravating Factors Positions functional movements postures or sustained loads that provoke or intensify the pain • Easing Factors Positions of relief rest thermal modalities or medications that alleviate the pain • **Severity** • Magnitude or intensity of the pain assessed at present at its best and at its worst over a defined period eg last 24 hours • **Nature of Pain** • Underlying clinical mechanism generating the symptoms • Mechanical Pain varies predictably with physical loads movements and rest • Inflammatory Constant aching marked morning stiffness warmth throbbing eased slightly by gentle movement • Neuropathic Radicular burning electric sensations with neurosensory deficits • Nociceptive / Central Sensitization Disproportionate pain intensity with hypersensitivity and non mechanical triggers • **Tissue Irritability** • Clinical concept Maitland determining how easily symptoms are provoked and how readily they settle dictating safe physical examination depth and treatment vigor • **Vigor/Threshold** The amount and intensity of activity required to bring on the pain • **Severity** The peak intensity of the provoked pain • **Latency / Time to Ease** The time required for pain to return to baseline once the provoking activity stops • **Triage** **High irritability** provoked

easily severe long time to ease requires gentle handling **Moderate irritability** moderate load needed moderate intensity eases in minutes **Low irritability** high load required mild pain eases immediately --- **Objective Measurement and Documentation** • **Visual Analogue Scale VAS** • Format A continuous horizontal unnumbered line exactly 100 mm 10 cm in length • Anchors Clearly demarcated endpoints labeled No pain 0 mm at the left and Worst pain imaginable 100 mm at the right • Administration The patient marks a single vertical tick across the line representing their current or peak pain level • Measurement and Documentation Measured with a millimeter ruler from the left anchor to the patients mark documented as a continuous value from 0 to 100 mm or 0 to 10 cm Highly sensitive to small clinical changes • **Numerical Rating Scale NRS** • Format An 11 point segmented numerical scale ranging from 0 to 10 • Anchors 0 equals No pain 1 to 3 equals Mild pain 4 to 6 equals Moderate pain 7 to 9 equals Severe pain 10 equals Worst possible pain • Administration Administered either verbally On a scale from 0 to 10 where 0 is no pain and 10 is the worst pain imaginable or visually as a numbered line • Documentation Recorded directly as a discrete integer score eg NRS 6/10 Highly practical for rapid clinical workflows and routine reassessment • **McGills Modified Questionnaire Short Form McGill Pain Questionnaire SF MPQ** • Format A multidimensional pain assessment instrument evaluating sensory affective and evaluative qualities rather than intensity alone • Sensory Component 11 sensory word descriptors throbbing shooting stabbing sharp cramping gnawing hot burning aching heavy tender splitting • Affective Component 4 affective word descriptors tiring exhausting sickening fearful punishing cruel • Scoring System Each of the 15 descriptors is rated by the patient on a 4 point intensity scale • 0 equals None • 1 equals Mild • 2 equals Moderate • 3 equals Severe • Indices Documented • Sensory Pain Rating Index S PRI Sum of sensory items 0 to 33 • Affective Pain Rating Index A PRI Sum of affective items 0 to 12 • Total PRI Combined sum 0 to 45 • Present Pain Intensity PPI A 6 point categorical index 0 equals No pain 1 equals Mild 2 equals Discomforting 3 equals Distressing 4 equals Horrible 5 equals Excruciating • Visual Analogue Scale VAS Standard 100 mm line included alongside the questionnaire for overall pain intensity

Investigations for Musculoskeletal Disorders BPT 3rd Year University Examination Notes --- 1 Blood Investigations and Markers • **Definition** • Blood investigations are laboratory tests performed to identify inflammation infection autoimmune disorders metabolic bone diseases and other abnormalities affecting the musculoskeletal system • **A Complete Blood Count CBC** • **Definition** A blood test that measures red blood cells white blood cells and platelets • **Purpose** • To detect infection and inflammation • To identify anemia associated with chronic diseases • **Important findings** • Increased WBC count May indicate infection • Decreased hemoglobin Indicates anemia • Increased platelet count May occur in inflammatory conditions • **B Erythrocyte Sedimentation Rate ESR** • **Definition** A test that measures the rate at which red blood cells settle in a tube over one hour • **Purpose** • To detect inflammation in the body • To help assess inflammatory and infectious musculoskeletal disorders • **Increased ESR may be seen in** • Rheumatoid arthritis • Osteomyelitis • Septic arthritis • Polymyalgia rheumatica • **C C Reactive Protein CRP** • **Definition** A protein produced by the liver in response to inflammation • **Purpose** • To detect acute inflammation • To monitor disease activity and response to treatment • **Increased CRP may be seen in** • Rheumatoid arthritis • Septic arthritis • Osteomyelitis • Other infections and inflammatory conditions • **D Rheumatoid Factor RF** • **Definition** An autoantibody commonly associated with rheumatoid arthritis • **Purpose** • To support the diagnosis of rheumatoid arthritis • **Positive RF may be seen in** • Rheumatoid arthritis • Other autoimmune diseases • Some chronic infections • **Note** A positive RF alone does not confirm rheumatoid arthritis • **E Anti Cyclic Citrullinated Peptide Anti CCP** • **Definition** An antibody test used to help diagnose rheumatoid arthritis • **Purpose** • To support early diagnosis of rheumatoid arthritis • To help identify patients at risk of more severe joint damage • **Important point** Anti CCP is relatively specific for rheumatoid arthritis • **F Serum Uric Acid** • **Definition** A blood test that measures the level of uric acid in the blood • **Purpose** • To support the evaluation of gout • **Increased uric acid may be seen in** • Gout • Hyperuricemia • **Note** A normal uric acid level does not rule out an acute gout attack • **G Serum Calcium and Phosphorus** • **Definition** Tests that measure calcium and phosphate levels in the blood • **Purpose** • To assess bone mineral metabolism • To investigate metabolic bone disorders • **Abnormal levels may be seen in** • Osteomalacia • Rickets • Hyperparathyroidism • Chronic kidney disease • **H Alkaline Phosphatase ALP** • **Definition** An enzyme found mainly in the liver and bones • **Purpose** • To assess bone formation and bone metabolism • **Increased ALP may be seen in** • Pagets disease of bone • Osteomalacia • Healing fractures • Increased bone turnover • **I Vitamin D Level** • **Definition** A blood test that measures 25 hydroxyvitamin D • **Purpose** • To assess vitamin D status • To investigate metabolic bone diseases • **Low vitamin D may be associated with** • Osteomalacia • Rickets • Reduced bone mineralization • **J Antinuclear Antibody ANA** • **Definition** A test that detects antibodies directed against components of the cell nucleus • **Purpose** • To help investigate systemic autoimmune diseases • **Positive ANA may be seen in** • Systemic lupus erythematosus • Other connective tissue diseases • **K HLA B27** • **Definition** A blood test that detects the HLA B27 genetic marker • **Purpose** • To support the diagnosis of spondyloarthritis • **Associated conditions** • Ankylosing spondylitis • Reactive arthritis • Other forms of spondyloarthritis • **Note** A positive result alone does not confirm the disease • **L Creatine Kinase CK** • **Definition** An enzyme found mainly in skeletal muscles heart and brain • **Purpose** • To detect muscle damage • To

investigate muscle diseases • **Increased CK may be seen in** • Muscular dystrophy • Myositis • Muscle injury • Rhabdomyolysis • **M Other Important Blood Investigations** • **Serum creatinine** Assesses kidney function especially before certain medications and contrast investigations • **Blood glucose** Helps assess diabetes which may affect musculoskeletal healing • **Blood cultures** Help identify the causative organism in suspected bone or joint infections --- **2 Radiological and Imaging Investigations** • **A X ray Plain Radiography** • **Definition** An imaging technique that uses X rays to produce images of bones and joints • **Principle** X rays pass through body tissues and are absorbed differently by different structures • **Indications** • To detect fractures and dislocations • To assess bone alignment and deformities • To identify joint space narrowing • To detect bone tumors and degenerative changes • **Advantages** • Quick and widely available • Relatively inexpensive • Useful for initial assessment of bone injuries • **Limitations** • Poor visualization of muscles ligaments and tendons • Uses ionizing radiation • Early bone and soft tissue abnormalities may not be visible • **B Computed Tomography CT Scan** • **Definition** A diagnostic imaging technique that uses X rays and computer processing to produce cross sectional images of the body • **Principle** Multiple X ray images are obtained from different angles and reconstructed into detailed cross sectional or 3D images • **Indications** • To assess complex fractures • To detect small bone fragments • To evaluate spinal fractures • To assess bone tumors • To assist in preoperative planning • **Advantages** • Provides detailed images of bones • Helps assess complex fractures • Can produce 3D images • **Limitations** • Uses ionizing radiation • Less effective than MRI for many soft tissue injuries • May require contrast in selected investigations • **C Magnetic Resonance Imaging MRI** • **Definition** An imaging technique that uses a strong magnetic field and radio waves to produce detailed images of body structures • **Principle** It detects signals from hydrogen nuclei in body tissues and converts them into detailed images • **Indications** • To detect ligament and tendon injuries • To assess intervertebral disc herniation • To identify meniscal injuries • To detect bone marrow abnormalities • To investigate soft tissue tumors and infections • **Advantages** • Excellent visualization of soft tissues • No ionizing radiation • Provides detailed images of muscles ligaments tendons and cartilage • **Limitations** • Expensive and less readily available • Takes longer than X ray or CT • Some implanted devices and metallic foreign bodies may make MRI unsuitable or require special precautions • **D Ultrasonography USG** • **Definition** An imaging technique that uses high frequency sound waves to produce images of body structures • **Principle** Sound waves are transmitted into the body and the reflected echoes are used to create images • **Indications** • To assess tendon injuries • To detect joint effusion • To identify muscle tears • To detect bursitis and soft tissue swelling • To guide joint aspiration and injections • **Advantages** • No ionizing radiation • Real time imaging • Allows dynamic assessment during movement • Relatively inexpensive and portable • **Limitations** • Highly dependent on operator skill • Limited visualization of deep structures • Cannot adequately assess bone marrow or structures hidden behind bone --- **3 Summary of Imaging Investigations** • **Investigation Main use** • X ray Fractures dislocations bone alignment • CT scan Complex fractures and detailed bone imaging • MRI Ligaments tendons cartilage discs and bone marrow • USG Tendons muscles effusions and dynamic assessment • **Clinical importance in physiotherapy** • Helps identify the underlying musculoskeletal pathology • Assists in selecting appropriate physiotherapy interventions • Helps identify contraindications and precautions • Supports monitoring of disease progression and recovery • Assists in determining the need for referral or further medical evaluation

--- **3 Modified Medical Research Council mMRC Dyspnea Scale** • **Definition** • The mMRC Dyspnea Scale is a subjective assessment tool used to grade breathlessness according to its effect on daily physical activities • **Purpose** • To assess the severity of dyspnea in respiratory disorders • To evaluate functional limitation during daily activities • To monitor disease progression and rehabilitation outcomes • Commonly used in patients with COPD and other chronic respiratory diseases • **Grades of Dyspnea** • **Grade 0** Breathlessness occurs only during strenuous exercise • **Grade 1** Breathlessness occurs when hurrying on level ground or walking up a slight hill • **Grade 2** Walks slower than people of the same age on level ground due to breathlessness or has to stop for breath while walking at own pace • **Grade 3** Stops for breath after walking approximately 100 meters or after a few minutes on level ground • **Grade 4** Too breathless to leave the house or becomes breathless while dressing or undressing • **Clinical Significance** • Higher grades indicate greater functional limitation due to breathlessness • Useful in assessing COPD severity and functional disability • Helps evaluate changes in symptoms during pulmonary rehabilitation • **Interpretation** • Grade 0 to 1 Minimal functional limitation • Grade 2 Moderate functional limitation • Grade 3 to 4 Severe functional limitation • Source National Academies of Sciences Engineering and Medicine • <https://www.ncbi.nlm.nih.gov/books/NBK593672/> --- **4 New York Heart Association NYHA Functional Classification** • **Definition** • The NYHA Functional Classification is a clinical assessment tool used to classify patients with cardiovascular disease according to the degree of limitation in physical activity caused by cardiac symptoms • **Purpose** • To assess functional capacity in patients with heart disease • To determine the severity of symptoms during physical activity • To monitor the patients clinical progress • Commonly used in patients with heart failure • **Grades of NYHA Classification** • **Class I** • No limitation of physical activity • Ordinary physical activity does not cause undue fatigue palpitation dyspnea or chest pain • **Class II** • Slight limitation of physical activity • Patient is comfortable at rest • Ordinary physical activity causes fatigue palpitation dyspnea or chest pain • **Class III** • Marked limitation of physical activity • Patient

is comfortable at rest • Less than ordinary physical activity causes fatigue palpitation dyspnea or chest pain • **Class IV** • Patient is unable to perform any physical activity without discomfort • Symptoms may be present even at rest • Any physical activity increases discomfort • **Clinical Significance** • Helps assess the functional status of patients with cardiovascular disorders • Useful in cardiac rehabilitation and exercise prescription • Helps monitor improvement or deterioration in functional capacity • Higher class indicates greater limitation in physical activity • Source American Heart Association • <https://www.heart.org/en/health-topics/heart-failure/what-is-heart-failure/classes-of-heart-failure> --- **5 Difference Between mMRC and NYHA Classification • mMRC Scale** • Primarily used to assess breathlessness related functional limitation in respiratory disorders • Consists of 5 grades 0 to 4 • Focuses mainly on breathlessness during daily activities • **NYHA Classification** • Primarily used to assess functional limitation in cardiovascular disorders especially heart failure • Consists of 4 classes I to IV • Focuses on symptoms such as dyspnea fatigue palpitation and chest pain during physical activity • Both scales are subjective and should be interpreted alongside clinical examination and other functional assessments

Exercise Tolerance Tests and Maximal Exercise Testing Protocols --- 1 12 Minute Walk Test 12MWT • Definition The 12 Minute Walk Test is a submaximal exercise test used to assess functional exercise capacity by measuring the maximum distance a person can walk in 12 minutes • **Purpose** • To assess cardiovascular and respiratory endurance • To evaluate functional exercise capacity • To monitor progress during pulmonary and cardiac rehabilitation • **Equipment** • Measured walking track • Stopwatch • Measuring tape • Pulse oximeter • Heart rate monitor • **Procedure** • Explain the test and demonstrate the walking technique • Record baseline heart rate blood pressure respiratory rate SpO₂ and perceived exertion • Ask the patient to walk as far as possible for 12 minutes • The patient may slow down stop or rest if required • Provide standardized encouragement • Record the total distance covered in 12 minutes • Monitor symptoms heart rate and SpO₂ during and after the test • **Outcome measures** • Total distance walked in meters • Heart rate response • Oxygen saturation • Dyspnea and fatigue using the Borg scale • **Advantages** • Simple inexpensive and easy to perform • Requires minimal equipment • Useful for assessing functional capacity • **Limitations** • Results are influenced by motivation walking speed and encouragement • Requires adequate space and patient ability to walk • **Clinical applications** • COPD and other respiratory disorders • Cardiac rehabilitation • Assessment of functional limitations --- **2 Shuttle Run Test • Definition** The Shuttle Run Test is a field based exercise test in which the individual runs or walks repeatedly between two marked points according to audio signals • **Purpose** • To assess cardiovascular endurance and aerobic fitness • To estimate maximal oxygen uptake VO₂ max • To evaluate exercise tolerance • **Types** • 20 Meter Multistage Shuttle Run Test • Incremental Shuttle Walk Test ISWT • **Equipment** • Flat measured track • Cones or markers • Audio recording and speaker • Stopwatch • Heart rate monitor and pulse oximeter when indicated • **Procedure** • Mark two points at the required distance apart • Explain and demonstrate the test • The individual moves between the markers according to audio beeps • The speed progressively increases in the multistage test • Record the number of shuttles or level completed • Monitor heart rate symptoms and perceived exertion • **Outcome measures** • Number of shuttles completed • Total distance covered • Final level or speed achieved • Estimated VO₂ max where validated for the specific protocol • **Advantages** • Requires relatively little equipment • Provides a standardized progressive workload • Useful for assessing aerobic fitness • **Limitations** • Requires adequate balance coordination and mobility • Results may be affected by motivation and musculoskeletal problems • Not suitable for all patients with severe cardiopulmonary disease • **Clinical applications** • Sports fitness assessment • Pulmonary rehabilitation • Assessment of aerobic exercise capacity --- **3 Step Test • Definition** The Step Test is a submaximal exercise test used to assess cardiovascular endurance by measuring the physiological response to stepping up and down on a platform at a standardized rate • **Purpose** • To assess cardiovascular endurance • To evaluate aerobic fitness • To assess heart rate recovery after exercise • To monitor fitness and rehabilitation progress • **Types** • Harvard Step Test • YMCA 3 Minute Step Test • Queens College Step Test • **Equipment** • Step platform of standardized height • Stopwatch or metronome • Heart rate monitor • Chair for recovery • **Procedure** • Record resting heart rate and blood pressure • Ask the subject to step up and down at a standardized rate • Maintain the required stepping cadence for the specified duration • Stop the test if the subject develops excessive fatigue dizziness chest pain or abnormal symptoms • Record heart rate during recovery • Calculate the fitness index or recovery score according to the selected protocol • **Outcome measures** • Heart rate response • Heart rate recovery • Duration of exercise completed • Fitness index depending on the protocol • **Advantages** • Simple and inexpensive • Requires minimal equipment • Useful for group fitness assessment • **Limitations** • Results depend on step height cadence and body weight • Requires adequate balance and lower limb strength • May not be suitable for patients with severe cardiopulmonary or musculoskeletal conditions • **Clinical applications** • Assessment of aerobic fitness • Fitness screening • Monitoring exercise training and rehabilitation --- **4 Treadmill Test • Definition** The Treadmill Test is an exercise test in which a person walks or runs on a treadmill while workload is progressively increased to assess cardiovascular and respiratory responses to exercise • **Purpose** • To assess cardiovascular endurance and exercise capacity • To evaluate heart rate and blood pressure responses • To detect exercise

induced ECG abnormalities • To assess symptoms such as chest pain dyspnea and fatigue • To estimate maximal oxygen uptake $\text{VO}_2 \text{ max}$ or metabolic equivalents METs • **Equipment** • Motorized treadmill • Electrocardiogram ECG machine • Blood pressure monitor • Heart rate monitor • Pulse oximeter when indicated • **Procedure** • Explain the test and obtain informed consent • Record resting heart rate blood pressure and ECG • Ask the subject to walk on the treadmill at the initial workload • Increase speed incline or both according to the selected protocol • Monitor ECG heart rate blood pressure symptoms and perceived exertion • Continue until maximal effort the target endpoint or a termination criterion is reached • Record recovery responses after stopping exercise • **Outcome measures** • Exercise duration • Maximum workload achieved • Heart rate and blood pressure response • ECG changes • METs or estimated $\text{VO}_2 \text{ max}$ • Symptoms and exercise tolerance • **Advantages** • Provides controlled and progressive exercise • Allows continuous ECG monitoring • Useful for evaluating cardiovascular function and exercise capacity • **Limitations** • Requires specialized equipment and trained personnel • May be unsuitable for individuals with balance or gait problems • Requires medical supervision when used for clinical cardiac stress testing • **Clinical applications** • Cardiac stress testing • Assessment of exercise tolerance • Cardiopulmonary rehabilitation • Evaluation of exercise induced symptoms --- **5 Theoretical Basis of Maximal Exercise**

Testing Protocols • **Definition** Maximal exercise testing protocols are standardized procedures in which exercise intensity is progressively increased until the individual reaches maximal effort or a clinically determined stopping point • **Purpose** • To assess maximal exercise capacity • To estimate or measure $\text{VO}_2 \text{ max}$ • To evaluate cardiovascular and respiratory responses to increasing workload • To identify exercise induced abnormalities • **Physiological basis** • During exercise oxygen demand of working muscles increases • Heart rate and stroke volume increase to raise cardiac output • Cardiac output equals Heart rate multiplied by Stroke volume • Pulmonary ventilation increases to meet oxygen demand and remove carbon dioxide • Oxygen uptake increases with workload until maximal oxygen uptake is reached or exercise is terminated • Workload is commonly expressed in METs • 1 MET equals 3 point 5 mL $\text{O}_2/\text{kg}/\text{min}$ representing a conventional estimate of resting oxygen consumption • **Principles of protocol design** • Workload should increase progressively • Initial workload should match the individuals functional capacity • Exercise duration should generally be sufficient to assess capacity without causing premature fatigue • The protocol should be selected according to age fitness clinical condition and testing purpose • ECG heart rate blood pressure and symptoms should be monitored during clinical stress testing • **Types of protocols** • Step protocols Workload increases at predetermined stages • Ramp protocols Workload increases continuously or in small increments • Fixed load protocols Workload remains constant throughout the test • **Common maximal exercise testing protocols** • Bruce Protocol • Modified Bruce Protocol • Balke Protocol --- **6 Bruce Protocol** • **Definition** The Bruce Protocol is a standardized maximal treadmill exercise test in which speed and treadmill incline increase every three minutes

• **Purpose** • To assess cardiovascular exercise capacity • To evaluate exercise induced ECG changes • To estimate maximal aerobic capacity • To assess functional capacity in individuals able to tolerate relatively high workloads • **Theoretical basis** • Uses progressive increases in speed and incline to increase exercise intensity • Each stage increases the metabolic demand placed on the cardiovascular and respiratory systems • The three minute stages allow physiological responses to develop before the next workload increase • Workload is expressed in METs to estimate functional capacity • **Procedure** • Record baseline ECG heart rate and blood pressure • Ask the subject to walk on the treadmill • Increase speed and incline every three minutes • Continue until maximal effort or a clinical termination criterion is reached • Monitor ECG heart rate blood pressure and symptoms throughout the test • **Standard Bruce Protocol stages** • Stage 1 1 point 7 mph 10 percent incline • Stage 2 2 point 5 mph 12 percent incline • Stage 3 3 point 4 mph 14 percent incline • Stage 4 4 point 2 mph 16 percent incline • Stage 5 5 point 0 mph 18 percent incline • Stage 6 5 point 5 mph 20 percent incline • Stage 7 6 point 0 mph 22 percent incline • **Advantages** • Widely used and standardized • Useful for assessing cardiovascular exercise capacity • Provides workload estimates in METs • **Limitations** • Initial workload and stage increments may be too high for sedentary or medically limited individuals • Rapid workload increases may cause early fatigue • Requires adequate walking ability and balance • **Clinical applications** • Exercise stress testing • Assessment of suspected coronary artery disease • Evaluation of functional exercise capacity --- **7 Modified Bruce Protocol** • **Definition** The Modified Bruce Protocol is a modified treadmill exercise test that includes additional low intensity stages before the standard Bruce stages • **Purpose** • To assess exercise capacity in individuals with reduced exercise tolerance • To evaluate patients who cannot tolerate the initial workload of the standard Bruce Protocol • To assess cardiovascular responses at lower exercise intensities • **Theoretical basis** • Provides a gradual increase in workload during the initial stages • Reduces the sudden increase in metabolic demand seen in the standard Bruce Protocol • Allows individuals with lower functional capacity to begin exercise at a more manageable intensity • **Procedure** • Record baseline ECG heart rate and blood pressure • Begin with low speed low incline treadmill walking • Increase workload in stages generally every three minutes • Continue into the standard Bruce stages as tolerated • Monitor ECG heart rate blood pressure and symptoms • **Modified Bruce Protocol stages** • Stage 1 1 point 7 mph 0 percent incline • Stage 2 1 point 7 mph 5 percent incline • Stage 3 1 point 7 mph 10 percent incline • Stage 4 2 point 5 mph 12 percent incline • Stage 5 3 point 4 mph 14 percent incline • Stage 6 4 point 2 mph 16 percent incline • Stage 7 5 point 0

mph 18 percent incline • **Advantages** • Suitable for individuals with lower exercise tolerance • Provides a more gradual initial workload progression • Allows assessment of patients who may not tolerate standard Bruce Stage 1 • **Limitations** • Requires treadmill equipment and monitoring • Test duration may be longer • Requires clinical supervision in patients undergoing cardiac stress testing • **Clinical applications** • Cardiac rehabilitation assessment • Patients with reduced functional capacity • Individuals requiring a lower initial exercise workload --- **8 Balke Protocol** • **Definition** The Balke Protocol is a maximal treadmill exercise test in which treadmill speed is maintained at a relatively constant level while the incline is progressively increased • **Purpose** • To assess maximal aerobic capacity • To evaluate cardiovascular endurance • To assess exercise capacity using a gradual increase in workload • **Theoretical basis** • Maintains a relatively constant walking speed • Progressively increases treadmill incline to increase exercise intensity • Produces a gradual increase in metabolic demand • Reduces the need for rapid changes in walking speed • **Procedure** • Record baseline heart rate blood pressure and ECG • Begin treadmill walking at a selected speed • Maintain the speed throughout the test • Increase the treadmill incline at regular intervals • Continue until maximal effort or a termination criterion is reached • Record exercise duration workload and physiological responses • **Common Balke Ware treadmill protocol** • Initial speed approximately 3 point 3 mph • Initial incline 0 percent • Incline is increased by approximately 1 percent each minute • Speed remains constant during the test • Variations exist according to the population and testing purpose • **Advantages** • Provides gradual workload increments • Useful for assessing aerobic capacity • Maintains a relatively constant walking speed • **Limitations** • Increasing incline may cause significant lower limb fatigue • Requires adequate walking ability • Different Balke modifications may produce different results • **Clinical applications** • Assessment of maximal aerobic capacity • Cardiopulmonary exercise testing • Evaluation of exercise tolerance --- **9 Comparison of Bruce Modified Bruce and Balke Protocols** • **Bruce Protocol** • Speed and incline increase every three minutes • Has relatively large workload increments • Used for individuals who can tolerate higher exercise intensity • **Modified Bruce Protocol** • Includes additional low intensity stages before the standard Bruce stages • Provides a more gradual initial increase in workload • Used for individuals with reduced exercise tolerance • **Balke Protocol** • Maintains a relatively constant treadmill speed • Progressively increases incline • Provides gradual workload progression --- **10 Safety Precautions During Maximal Exercise Testing** • Obtain relevant medical history and assess contraindications before testing • Record baseline heart rate blood pressure and ECG when clinically indicated • Ensure appropriate supervision and emergency equipment during clinical stress testing • Monitor heart rate blood pressure ECG symptoms and perceived exertion • Stop the test if the individual develops chest pain severe breathlessness dizziness abnormal ECG changes a concerning blood pressure response or other significant symptoms • Provide an appropriate recovery period and monitor the patient after exercise --- **References** • American Thoracic Society/American College of Chest Physicians ATS/ACCP Statement on Cardiopulmonary Exercise Testing • Treadmill Stress Testing StatPearls NCBI Bookshelf • Cardiopulmonary Exercise Testing in Congenital Heart Disease Equipment and Test Protocols